
Fundamentals of Information Systems Security

Fourth Edition

Project 4

Penetration Testing Report

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Date: _____

Weight: 20%

Submission Format: PDF Report via Course Platform

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1 Project Overview

About This Project

In this project, you will conduct a simulated penetration test on a target system (Metasploitable 2 or a similar vulnerable VM) and produce a professional penetration testing report. You will apply the skills learned in Labs 2, 4, 6, and 7 to identify vulnerabilities, exploit them, and document findings.

The report should follow industry standards for penetration testing reports (e.g., PTES, OWASP, or a simplified version).

Important Ethical Notice

This is a simulated exercise. Do not perform any actual attacks on real systems without explicit written authorization. All testing must be conducted in the provided lab environment (Metasploitable 2). Unauthorized access is illegal and unethical.

2 Learning Objectives

Upon completion of this project, you will be able to:

1. Plan and execute a penetration test following a structured methodology
2. Use various security tools (Nmap, Metasploit, Wireshark, etc.)
3. Identify and exploit vulnerabilities in a target system
4. Document findings clearly and professionally
5. Provide actionable remediation recommendations

3 Scenario

Organization Profile

You have been hired as an external penetration tester by **SecureTech Ltd.**, a cloud service provider. They have provided you with a test environment (Metasploitable 2) to assess the security of their internal network. You must conduct a controlled penetration test, identify as many vulnerabilities as possible, and produce a comprehensive report.

4 Project Tasks

4.1 Task 1: Reconnaissance and Information Gathering

Task 1 Instructions

1. Perform network scanning to identify live hosts, open ports, and services
2. Enumerate services (FTP, SSH, HTTP, SMB, etc.) and versions
3. Document the target's network topology

4.2 Task 2: Vulnerability Analysis

Task 2 Instructions

1. Identify potential vulnerabilities using Nmap scripts, manual checks, and vulnerability databases (CVE)
2. Prioritize vulnerabilities based on severity and exploitability
3. Create a vulnerability list

4.3 Task 3: Exploitation

Task 3 Instructions

1. Exploit at least three different vulnerabilities (e.g., vsftpd backdoor, UnrealIRCd, Samba, Distcc, etc.)
2. Use Metasploit or manual exploitation techniques
3. Gain access (shell or Meterpreter) and demonstrate privilege escalation if possible

4.4 Task 4: Post-Exploitation and Evidence

Task 4 Instructions

1. Gather system information (users, files, network configuration)
2. Capture screenshots of successful exploits and commands
3. Document the steps taken for each exploitation

4.5 Task 5: Reporting

Task 5 Instructions

1. Write a structured penetration testing report
2. Include findings, evidence, risk ratings, and remediation steps

5 Deliverables

You must submit a professional penetration testing report containing:

1. **Executive Summary** (for management)
2. **Scope and Methodology**
3. **Reconnaissance Results** (network map, open ports, services)
4. **Vulnerability Assessment** (list of vulnerabilities with risk ratings)
5. **Exploitation Summary** (description and screenshots of successful exploits)
6. **Post-Exploitation Findings** (data accessed, privileges obtained)
7. **Remediation Recommendations** (prioritized)
8. **Technical Appendix** (detailed steps, logs, commands)

6 Evaluation Rubric

Criteria	Weight	Description
Reconnaissance and Enumeration	15%	Thoroughness and accuracy of scanning and enumeration
Vulnerability Identification	15%	Comprehensive vulnerability list with proper prioritization
Exploitation Success	25%	Successful exploitation of at least three vulnerabilities
Evidence and Screenshots	15%	Clear, annotated screenshots for each major step
Report Quality and Professionalism	15%	Well-structured, well-written, following industry standards
Remediation Recommendations	10%	Feasible, actionable recommendations
Critical Thinking	5%	Demonstrates understanding of the attack chain and risk context

7 Submission Guidelines

Submission Requirements

- Submit a single PDF file via the course platform
- File name: `Project4_PenTest_YourName.pdf`
- Include a cover page, table of contents, and section numbering
- All screenshots must be labelled and referenced in the text

8 Due Date and Late Policy

Important Deadlines

- **Due Date:** _____ (Week 13)
- **Late Policy:** Submissions will be penalized 10% per day
- **Extensions:** Must be requested at least 48 hours in advance

9 Resources

- Lab guides (Labs 2, 4, 6, 7)
- Metasploitable 2 VM
- Nmap, Metasploit, Wireshark documentation
- PTES (Penetration Testing Execution Standard) and OWASP Testing Guide